Data Meter

Faculty: Prof. Chakradhar Saswade

By: Poorvesh Dali

Assignment 1

Thorpe tube

A Thorpe tube is an analogue variable-area flowmeter used universally in clinical settings, ambulances, and portable oxygen systems to measure and control the flow rate of medical gases

It is entirely mechanical and fail-safe. It requires zero electrical power, batteries, or software. In an emergency blackout or ambulance crash, as long as there is gas pressure, the tube accurately displays flow.



UX Audit

Reading a floating ball inside a cylinder requires exact eye-level alignment. Viewing the tube from an angle common in cramped hospital rooms or moving ambulances—frequently causes dosing misreads of up to ±1 LPM.

It fails to answer the user's primary question: "How much time do I have left before this tank runs out?"

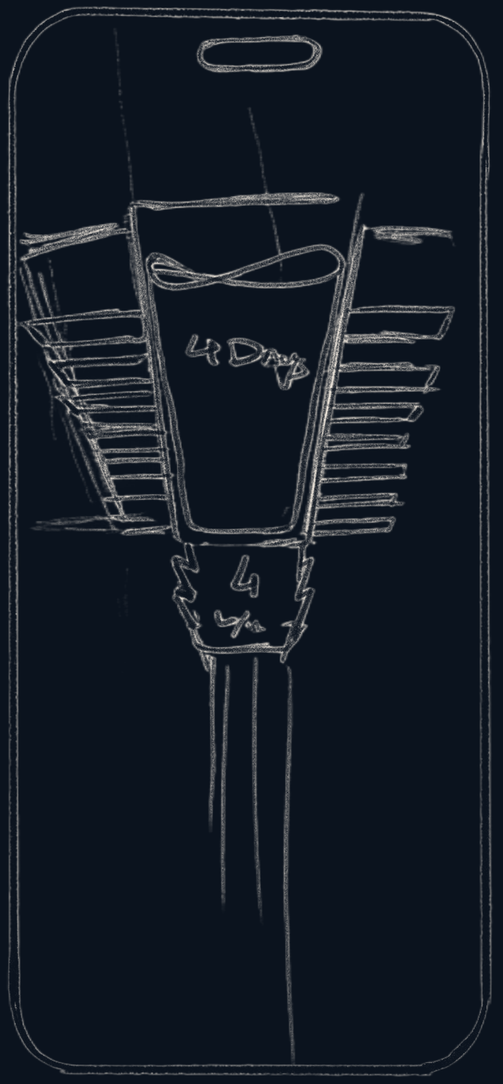
At night, with low light it’s difficult to see the value

Ideation

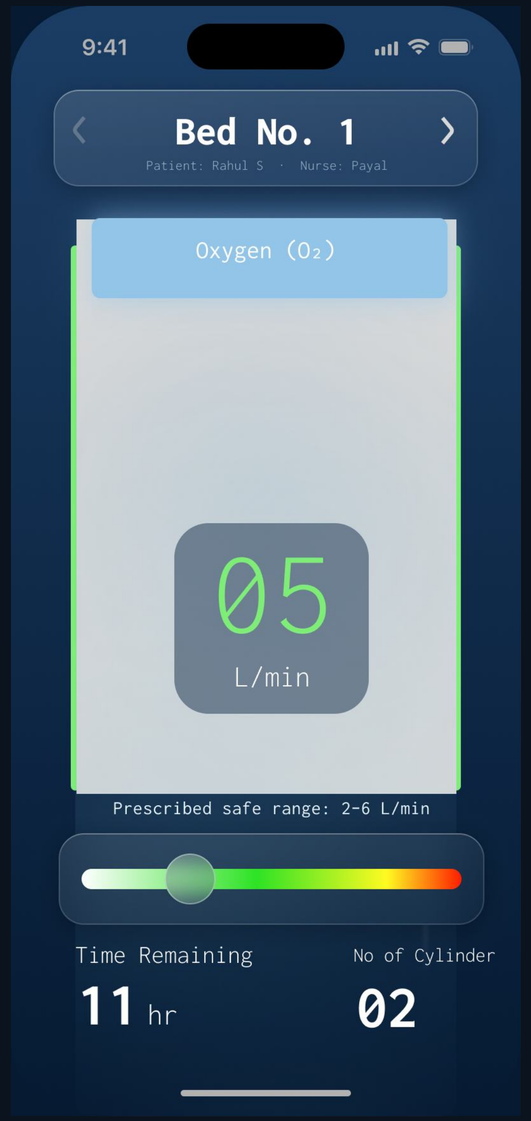
Initially some ideas were explored, considering the research and UX audit







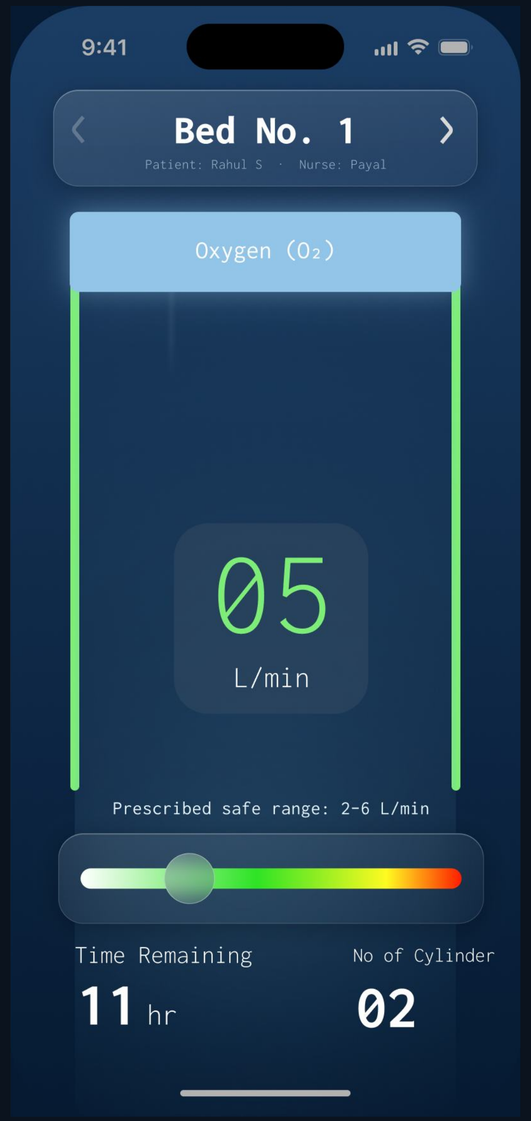
Exploration



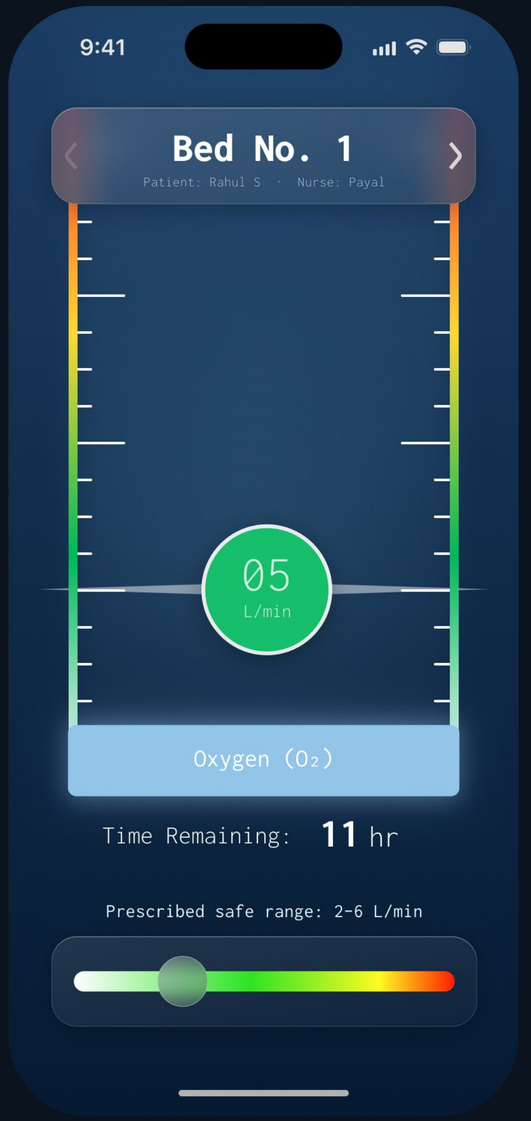
Exploration 1



Exploration 2

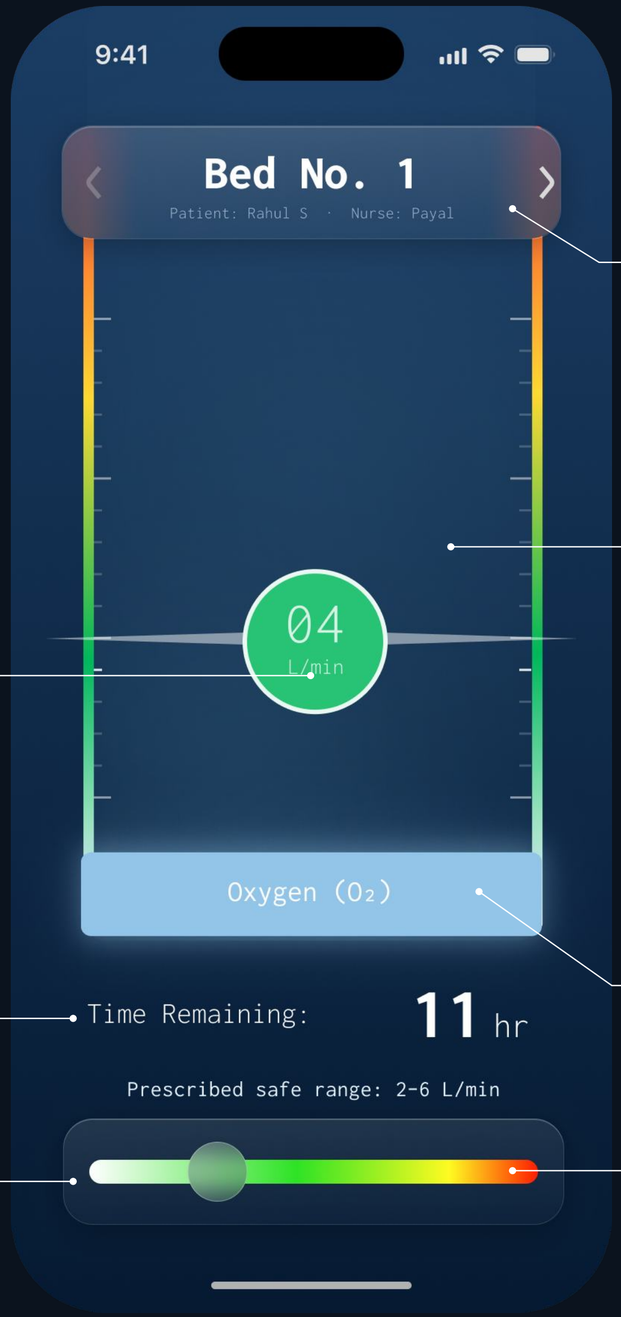


Exploration 3



Exploration 4

UI Details



Patience details:
- Bed no.
- Patient name
- Nurse name

Flow Rate Value

Air flowing animation to show the flow rate intensity

Gas details

Time Remaining for gas to finish

Slide to change the value

Gradient to show the intensity